

OTEPIC

LAND PREPARATION & SEEDBED PREPARATION



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TRAINING MANUAL HANDBOOK

LAND/BED PREPARATION

LAND PREPARATION

Double Digging

It is gardening technique that involves loosening and mixing organic matter into two layers of soil to improve drainage and aeration

How to double Dig

If you do decide your soil would benefit from double digging, follow these steps:

1. First, dig a trench with a spade along the one side of a bed. The soil from this trench is set aside.
2. Fork over the base of the trench using a garden fork. Ideally, incorporate organic matter such as garden compost at this stage – use about a bucketful per square meter
3. Dig the next trench. Take a step backwards, creating a gap between your feet and the first trench – in this gap, dig your next trench alongside the first (parallel). The soil dug from this second trench is turned into the first trench (filling it back up).
4. Repeat step 3 until the whole area is dug.
5. When you reach the end, the soil set aside from the first trench is used to fill the last.

Soil preparation where double digging is required or advisable

- ✓ Cultivating soil in new gardens, where the builders may not have left you much top soil
- ✓ where you have a heavy, unworkable clay soil
- ✓ where you need to improve the nutrients of a soil by incorporating organic matter at root level rather than as a mulch
- ✓ creating new borders; perhaps in an area that has been lawn
- ✓ if you're planning an orchard or even a few fruit trees planted together in one area; a good deep layer of top soil is useful
- ✓ where you have invasive, perennial weeds such as bindweed or ground elder; although there are likely to be some weeds remaining, it's a more thorough weeding practice than just digging them up individually.
- ✓ on an allotment, where it hasn't been cultivated for a while
- ✓ regenerating a perennial or mixed herbaceous flower border

Advantages of soil preparation using double digging

It will: –

1. Increase the amount of space in the soil for air and water, which in turn improves plants growth.
2. Improve the soil structure, especially when compost or manure are added.
3. Promote earthworms and the activities of soil microbes.
4. Break up soil crusts and hard pans, and increases the speed that water that can seep into the soil.
5. Help the soil retain water and nutrients when used with organic fertilizer

PREPARING A LAZY BED

A seedbed or seedling bed is the local soil environment in which seeds are planted. The key to a productive, healthy garden is the preparation of the growing beds. A well-prepared bed with loose soil to a depth of 24 inches allows the roots of the plants to grow evenly and to provide a steady supply of nutrients to the rest of the plant. Water is able to move through the soil freely, and weeds are easy to pull out. The plant roots have so much loose soil to grow into that more plants grow in an even area, and this means more food from a smaller garden.



Tools that are required in starting a new bed

1. *Hoe*
2. *Plank for standing (roughly 6ft or 2m) Measuring string*
3. *Good wooden stakes of approximately 75cm*

The preparation of a seedbed may include:

1. **The removal of weeds.** They contain Insect eggs and disease spores and so this has to be removed from the plot. Stones will also physically prevent the seedlings from growing and so must be removed too.



2. **Levelling.** The site will have been levelled for even drainage.



3. **Breaking up the soil.** Compacted soil will have to be broken up by digging to allow air and water circulation, this helps the seedling penetrate the soil.



4. **Soil improvement.** The soil structure can improved by watering and introduction of organic matter such as compost or peat.



5. **Planting** – Plant the seedlings with reasonable spacing to allow aeration

TIPS: Avoid Recompressing the soil. Once the bed has been dug, try not to walk on it. One of the reasons for double digging is to put air in the soil. Walking on the bed will re-compact the soil. When planting the seedlings in the bed, using a digging board will allow you to distribute your weight over a wider area and minimize compaction.